Assignment 9

Topic: Responsive Design, Flexbox, Grid & Media Queries

Q1. Responsive Image Filter Gallery

RESPONSIVE-IMAGE-FILTER-GALLERY.HTML

```
<!DOCTYPE html>
<html lang="en">
<head><link rel="stylesheet" href="responsive-image-filter-gallery.css"></head>
<body>
    <header><h1>Filter Gallery</h1></header>
    <main class="gallery">
        <figure><img src="sekiro.jpg" class="filter-blur"><figcaption>Blur</
figcaption></figure>
        <figure><img src="cod.jpg" class="filter-grayscale"><figcaption>Grayscale</
figcaption></figure>
        <figure><img src="img3.jpg" class="filter-sepia"><figcaption>Sepia</
figcaption></figure>
        <figure><img src="img4.jpg" class="filter-
brightness"><figcaption>Brightness</figcaption></figure>
        <figure><img src="img5.jpg" class="filter-contrast"><figcaption>Contrast</
figcaption></figure>
        <figure><img src="img6.jpg" class="filter-hue"><figcaption>Hue Rotate</
figcaption></figure>
    </main>
</body>
</html>
```

RESPONSIVE-IMAGE-FILTER-GALLERY.CSS

```
body { font-family: sans-serif; margin: 0; padding: 20px; text-align: center; }
.gallery { display: grid; grid-template-columns: repeat(auto-fit, minmax(200px,
1fr)); gap: 20px; }
figure { margin: 0; overflow: hidden; border-radius: 8px; }
img { width: 100%; height: auto; display: block; transition: filter 0.4s ease; }
figcaption { padding: 10px; background: #333; color: #fff; }
/* Filters */
.filter-blur { filter: blur(5px); }
.filter-grayscale { filter: grayscale(100%); }
.filter-sepia { filter: sepia(100%); }
.filter-brightness { filter: brightness(150%); }
.filter-contrast { filter: contrast(200%); }
.filter-hue { filter: hue-rotate(90deg); }
/* Hover removes filter */
img:hover { filter: none; }
/* Mobile Stack */
@media (max-width: 600px) {
```

```
    .gallery { grid-template-columns: 1fr; }
}
```

Q2. Responsive Web Design Layout

RESPONSIVE-WEB-DESIGN-LAYOUT.HTML

```
<body>
    <header class="site-header">
        <div class="logo">MyLogo</div>
        <nav><a href="#">Home</a> <a href="#">About</a></nav>
    </header>
    <main class="features">
        <section class="feature-box">Feature 1</section>
        <section class="feature-box">Feature 2</section>
        <section class="feature-box">Feature 3</section>
    </main>
    <footer class="site-footer"><p>Centered Footer</p></footer>
</body>
```

RESPONSIVE-WEB-DESIGN-LAYOUT.CSS

```
body { margin: 0; font-family: sans-serif; display: flex; flex-direction: column;
min-height: 100vh; }
.site-header { display: flex; justify-content: space-between; padding: 20px;
background: #2c3e50; color: white; }
.site-header nav a { color: white; text-decoration: none; margin-left: 15px; }
.features { display: flex; justify-content: center; gap: 20px; padding: 40px; flex:
1; }
.feature-box { background: #ecf0f1; padding: 40px; flex: 1; text-align: center;
transition: transform 0.3s; border-radius: 8px; }
.feature-box:hover { transform: scale(1.05); }
.site-footer { text-align: center; padding: 20px; background: #34495e; color:
white; }
@media (max-width: 750px) {
    .features { flex-direction: column; }
}
```

Q3. Responsive Cards with Media Queries

RESPONSIVE-CARDS-WITH-MEDIA-QUERIES.HTML

```
<main class="card-row">
    <div class="responsive-card">Card A</div>
    <div class="responsive-card">Card B</div>
    <div class="responsive-card">Card C</div>
    <div class="responsive-card">Card D</div>
</main>
```

RESPONSIVE-CARDS-WITH-MEDIA-QUERIES.CSS

```
.card-row { display: flex; flex-wrap: wrap; gap: 20px; padding: 20px; }
.responsive-card {
    flex: 0 0 calc(25% - 15px); /* 4 columns desktop */
    background: #fff; border: 1px solid #ddd; padding: 30px; text-align: center;
    box-sizing: border-box; transition: background-color 0.3s; border-radius: 6px;
}
.responsive-card:hover { background-color: #3498db; color: white; }
@media (max-width: 900px) {
    .responsive-card { flex: 0 0 calc(50% - 10px); } /* 2 columns tablet */
}
@media (max-width: 600px) {
    .responsive-card { flex: 0 0 100%; } /* 1 column mobile */
}
```

Q4. Responsive Typography

RESPONSIVE-TYPOGRAPHY.HTML

```
<section class="typo-section">
    <h1>Fluid Typography Demo</h1>
    <p>This text scales seamlessly based on the viewport width using the CSS clamp()
function, ensuring perfect readability across all devices.</p>
</section>
```

RESPONSIVE-TYPOGRAPHY.CSS

```
.typo-section { text-align: center; margin: 0 auto; padding: 50px 20px; max-width:
800px; }
.typo-section h1 { font-size: clamp(1.5rem, 4vw, 3rem); color: #2c3e50; }
.typo-section p { font-size: clamp(1rem, 2vw, 1.5rem); color: #555; line-height:
1.6; }
@media (max-width: 700px) {
    .typo-section { padding: 30px 15px; }
    .typo-section p { line-height: 1.4; }
}
```

Q5. Filter Cards with Animation

FILTER-CARDS-WITH-ANIMATION.HTML

```
<main class="profiles">
    <div class="profile-card"><img src="nitin.jpg" alt="Profile"><h3>Nitin (Front-End)</h3></div>
    <div class="profile-card"><img src="manthan.jpg" alt="Profile"><h3>Manthan</h3></div>
    <div class="profile-card"><img src="user3.jpg" alt="Profile"><h3>Developer</h3></div>
</main>
```

FILTER-CARDS-WITH-ANIMATION.CSS

```
@keyframes fadeIn { from { opacity: 0; transform: scale(0.95); } to { opacity: 1; transform: scale(1); } }
.profiles { display: flex; justify-content: center; gap: 30px; padding: 40px; animation: fadeIn 1s ease-out; }
.profile-card { background: white; padding: 20px; border-radius: 10px; text-align: center; box-shadow: 0 4px 6px rgba(0,0,0,0.1); transition: transform 0.3s; }
.profile-card img { width: 100%; max-width: 150px; border-radius: 50%; filter: sepia(80%); transition: filter 0.3s; }
.profile-card:hover { transform: translateY(-10px); }
.profile-card:hover img { filter: none; }
@media (max-width: 800px) {
    .profiles { flex-direction: column; align-items: center; }
}
```

Q6. Responsive CSS Cheat Sheet

RESPONSIVE-CSS-CHEAT-SHEET.HTML

```
<div class="cheat-sheet">
    <article class="cheat-item"><h3>Filters</h3><p>Visual effects like blur and
brightness applied to elements.</p></article>
    <article class="cheat-item"><h3>Media Queries</h3><p>Applies CSS rules based on
device breakpoints (e.g., max-width).</p></article>
    <article class="cheat-item"><h3>Responsive Design</h3><p>Creating layouts that
adapt seamlessly to any screen size.</p></article>
    <article class="cheat-item"><h3>Typography</h3><p>Using relative units (vw, rem)
and clamp() for fluid fonts.</p></article>
</div>
```

RESPONSIVE-CSS-CHEAT-SHEET.CSS

```
.cheat-sheet { display: grid; grid-template-columns: repeat(auto-fit, minmax(300px,
1fr)); gap: 20px; padding: 30px; }
.cheat-item { background: #fdfdfd; border-left: 5px solid #e74c3c; padding: 20px;
box-shadow: 0 2px 5px rgba(0,0,0,0.1); transition: background 0.3s, transform
0.3s; }
.cheat-item:hover { background: #ffeaa7; transform: translateX(5px); }
.cheat-item h3 { margin-top: 0; color: #c0392b; }
@media (max-width: 700px) {
    .cheat-sheet { display: flex; flex-direction: column; }
    .cheat-item { display: flex; justify-content: space-between; align-items:
center; }
    .cheat-item h3 { margin: 0; margin-right: 15px; }
    .cheat-item p { margin: 0; text-align: right; font-size: 0.9em; }
}
```

Q7. Combined Responsive Web Demo

COMBINED-RESPONSIVE-WEB-DEMO.HTML

```
<body class="demo-page">
    <header class="flex-header"><h1>Responsive Demo</h1><nav><a href="#">Link</a></
nav></header>
    <main class="grid-main">
        <section class="demo-gallery">Gallery Area</section>
        <section class="demo-text">Explanation of responsive layout goes here.</
section>
        <section class="demo-preview">Cheat Sheet Preview</section>
    </main>
    <footer class="demo-footer">Contact: admin@example.com</footer>
</body>
```

COMBINED-RESPONSIVE-WEB-DEMO.CSS

```
@keyframes fade { from { opacity: 0; } to { opacity: 1; } }
.demo-page { margin: 0; display: flex; flex-direction: column; min-height: 100vh;
font-family: sans-serif; animation: fade 1.2s ease-in; }
.flex-header { display: flex; justify-content: space-between; background: #34495e;
padding: 20px; color: white; }
.grid-main { display: grid; grid-template-columns: 1fr 2fr 1fr; gap: 20px; padding:
20px; flex: 1; }
.grid-main section { background: #ecf0f1; padding: 20px; border-radius: 5px;
transition: transform 0.3s; }
.grid-main section:hover { transform: scale(1.02); }
.demo-footer { text-align: center; padding: 15px; background: #2c3e50; color:
white; }
@media (max-width: 700px) {
    .grid-main { grid-template-columns: 1fr; }
}
```

Q8. Filter Comparison Layout

FILTER-COMPARISON-LAYOUT.HTML

```
<main class="comparison-container">
    <figure><img src="original.jpg" alt="Original"><figcaption>Before</figcaption></
figure>
    <figure><img src="filtered.jpg" class="apply-filter"
alt="Filtered"><figcaption>After (Hover to Reveal)</figcaption></figure>
</main>
```

FILTER-COMPARISON-LAYOUT.CSS

```
.comparison-container { display: flex; justify-content: center; gap: 20px; padding:
40px; }
.comparison-container figure { margin: 0; flex: 1; max-width: 500px; }
.comparison-container img { width: 100%; display: block; border-radius: 8px; }
.apply-filter { filter: brightness(150%) contrast(120%); transition: filter 0.5s
ease; }
.apply-filter:hover { filter: none; }
figcaption { text-align: center; margin-top: 10px; font-weight: bold; }
@media (max-width: 700px) {
    .comparison-container { flex-direction: column; align-items: center; }
}
```

Q9. Responsive Design Principles Summary

RESPONSIVE-DESIGN-PRINCIPLES-SUMMARY.HTML

```
<div class="principles-grid">
    <div class="principle-box">
        <h3>Fluid Layouts</h3>
        <p>Use relative units like % or vw instead of px.</p>
        <code>width: 100%;</code>
    </div>
    <div class="principle-box">
        <h3>Flexible Media</h3>
        <p>Ensure images scale within their containers.</p>
        <code>max-width: 100%; height: auto;</code>
    </div>
    <div class="principle-box">
        <h3>Breakpoints</h3>
        <p>Apply styles based on screen width.</p>
        <code>@media (max-width: 600px)</code>
    </div>
</div>
```

RESPONSIVE-DESIGN-PRINCIPLES-SUMMARY.CSS

```
.principles-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 20px;
padding: 30px; }
.principle-box { background: #eee; padding: 20px; border-radius: 8px; text-align:
center; transition: all 0.3s ease; }
.principle-box code { display: block; margin-top: 15px; background: #333; color:
#0f0; padding: 10px; border-radius: 4px; }
.principle-box:hover { background: linear-gradient(135deg, #6a11cb 0%, #2575fc
100%); color: white; }
@media (max-width: 600px) {
    .principles-grid { grid-template-columns: 1fr; }
}
```

Q10. Responsive Dashboard Summary

RESPONSIVE-DASHBOARD-SUMMARY.HTML

```
<div class="dashboard">
    <header class="dash-header">Dashboard Header</header>
    <aside class="dash-sidebar">Sidebar Nav</aside>
    <main class="dash-content">
        <div class="dash-card">Filters</div>
        <div class="dash-card">Typography</div>
        <div class="dash-card">Media Queries</div>
        <div class="dash-card">Cheat Sheet Links</div>
    </main>
</div>
```

RESPONSIVE-DASHBOARD-SUMMARY.CSS

```
@keyframes slideIn { from { transform: translateY(20px); opacity: 0; } to
{ transform: translateY(0); opacity: 1; } }
.dashboard { display: grid; grid-template-areas: "header header" "sidebar main";
grid-template-columns: 250px 1fr; grid-template-rows: 60px 1fr; min-height: 100vh; }
.dash-header { grid-area: header; background: #2c3e50; color: white; display: flex;
align-items: center; padding: 0 20px; }
.dash-sidebar { grid-area: sidebar; background: #34495e; color: white; padding:
20px; }
.dash-content { grid-area: main; display: grid; grid-template-columns: repeat(auto-
fit, minmax(200px, 1fr)); gap: 20px; padding: 20px; background: #ecf0f1; animation:
slideIn 0.8s ease-out; }
.dash-card { background: white; padding: 30px; border-radius: 8px; text-align:
center; box-shadow: 0 2px 4px rgba(0,0,0,0.05); transition: transform 0.3s, box-
shadow 0.3s; }
.dash-card:hover { transform: scale(1.05); box-shadow: 0 8px 15px rgba(0,0,0,0.1); }
@media (max-width: 750px) {
    .dashboard { grid-template-areas: "header" "sidebar" "main"; grid-template-
columns: 1fr; grid-template-rows: auto auto 1fr; }
}
```